

James Friedenber

Software Engineer | Full-Stack Engineer & Platform Systems

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[LinkedIn](#) | [Portfolio](#) | [Github](#)

SUMMARY

Software Engineer with experience building scalable live-event systems, developer tooling, client/server performance infrastructure, and modern web applications at Microsoft. Directly supported millions of players, large-scale live events, and creator ecosystems.

PROFESSIONAL EXPERIENCE

Software Engineer, Microsoft - Remote, USA

March 2022 - June 2026

- Co-engineered Minecraft: Bedrock Edition's first automated multiplayer performance testing framework, increasing player-per-server capacity from 10 to 80 and reducing infrastructure requirements for a 25k-player live event from ~2,500 to ~313 Azure VMs.
- Developed foundational gameplay and event systems, including Minecraft's first official in-game cutscene system, for the Minecraft Legends 2023 promotional live event, reaching over 3M players.
- Rebuilt Minecraft live events management tooling as a React web application, integrating it into the Auger website content pipeline to enable distribution to third party creators and streamline the live events deployment process.
- Designed and implemented a telemetry and observability framework for MCTools using Azure Application Insights, enabling custom event instrumentation, real-time monitoring, issue detection, and operational visibility across the platform.
- Developed a localization framework, enabling MCTools to be available in 20+ languages and bringing it into compliance with the [minecraft.net](#) domain.
- Modernized the MCTools TypeScript codebase and expanded content validation suites to reduce manual review workload for the Minecraft Vanilla team and external creators.
- Mentored and onboarded new engineers to the Minecraft web and platform technology stack, providing guidance on React, TypeScript, deployment workflows, telemetry tooling, and development practices.

Software Design Engineer in Test, DIGARC - Remote, USA

March 2021 - March 2022

- Developed detailed test plans and tracked quality initiatives using Jira for an Agile team of developers, testers, and a PM.
- Created and maintained several hundred automated E2E tests, ensuring reliable frontend regression coverage.
- Led weekly team meetings to evaluate performance and functionality metrics in preparation for production deployments.

Software Design Engineer in Test, Andromeda Systems Inc. - Jacksonville, Florida

August 2019 - March 2021

- Developed over 500 automated tests using Selenium for an inventory management application used by the U.S. Navy for aircraft maintenance.
- Built dozens of Angular components using TypeScript for administrative tools to improve client operations and workflows.

- Ensured product quality in an Agile environment by triaging bugs and writing user stories and test cases.

QA Automation Engineer CO-OP, LenelS2 - Rochester, New York

January 2018 - August 2018

- Developed and designed a cloud-based facilities security product, contributing Selenium frontend automated tests and UI/UX improvements.
- Tested software interactions with security scanners by configuring and running automated tests in Linux terminals.
- Ensured product specifications met security standards and maintained quality by developing detailed test plans in an Agile environment.

PERSONAL PROJECTS

Personal Website Platform

January 2026 - Present

- Designed and deployed a full-stack personal web platform using React, TypeScript, Node.js, and Azure cloud services.
- Implemented backend service hosting, environment configuration, API deployment, and PostgreSQL integration on Azure.
- Established CI/CD workflows for automated deployment and infrastructure management.
- Configured application monitoring, logging, and operational tooling to support ongoing maintenance and troubleshooting.
- Managed the complete deployment lifecycle independently, including cloud provisioning, configuration, deployment, and production support.

SKILLS

Languages: TypeScript/JavaScript, C++, C#, Java, Python

Frontend: React, Angular, TailwindCSS, Zod, Zustand

Backend and Architecture: Node.js API Development, .NET 8+, Cloud Deployment, Telemetry & Monitoring

Cloud and Infra: Git, Azure App Services, Azure PostgreSQL, Azure Application Insights, CI/CD Pipelines, Environment Configuration Management, REST API Deployment, SQL

Testing and Automation: Playwright, Selenium, React Testing Library, Jest, Postman, Swagger

AI-Assisted Development: GitHub Copilot, Large Language Models, Generative AI, OpenAI API, Retrieval-Augmented Generation, Vector Databases

EDUCATION

Rochester Institute of Technology - BS in Game Design and Development

August 2015 - May 2019